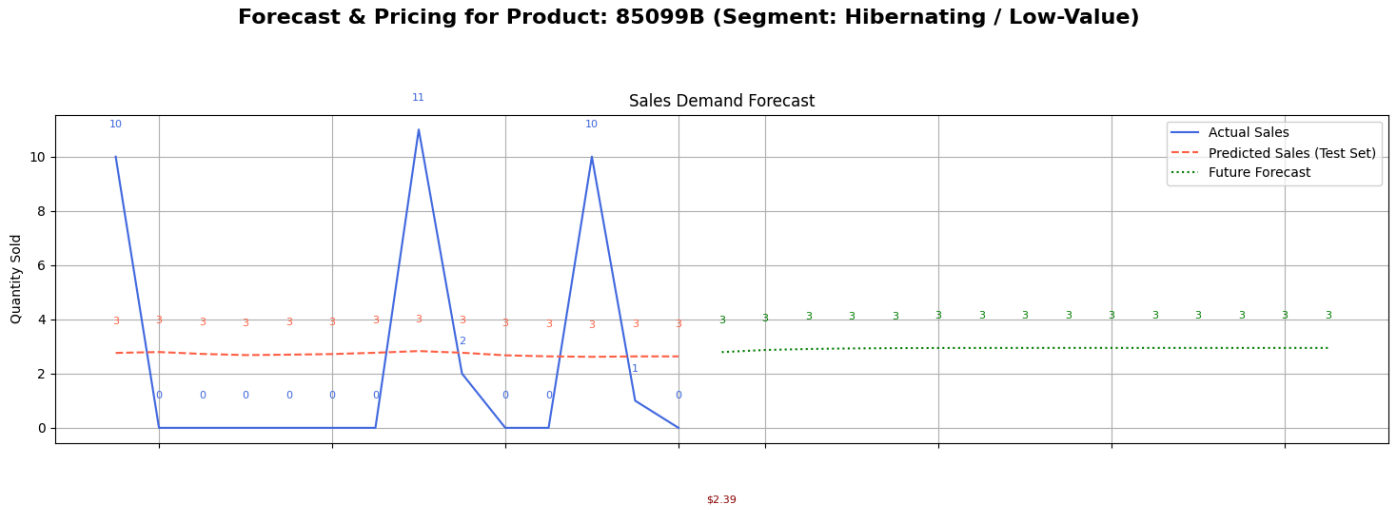
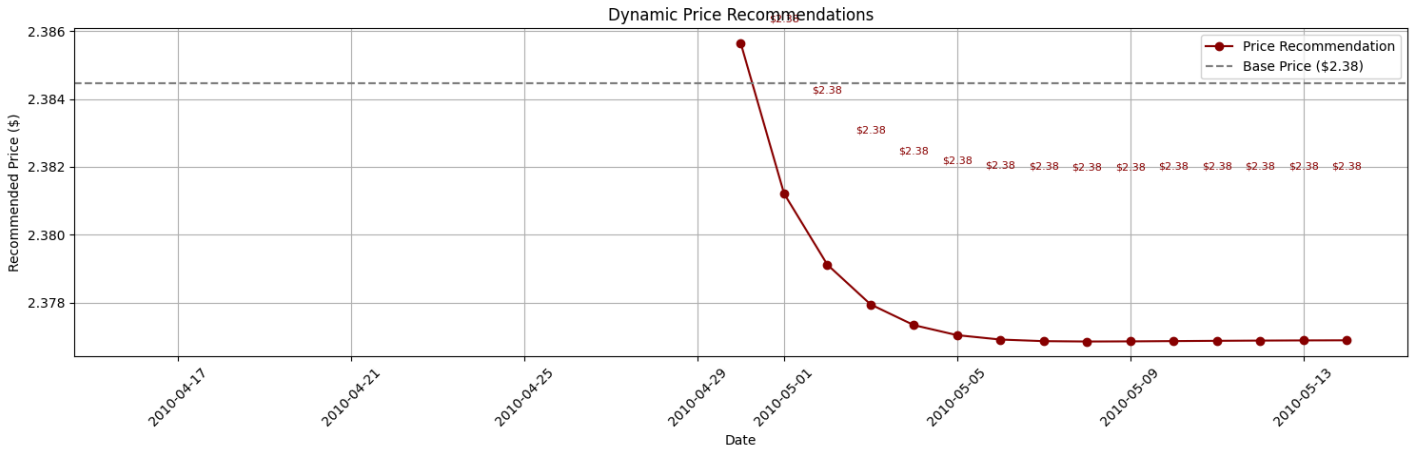
```
============================================================
--- Starting LSTM Forecast Pipeline for Product: 85099B, Segment: Hibernating / Low-Value ---
============================================================

--- Preparing data for Segment: Hibernating / Low-Value ---
-> Data prepared successfully for 'Hibernating / Low-Value' with 150 daily records.
--- Training model for Segment: Hibernating / Low-Value ---
Epoch 10/100, Val Loss: 0.0247
Epoch 20/100, Val Loss: 0.0259
Epoch 30/100, Val Loss: 0.0456
Early stopping at epoch 30.
-> Model for 'Hibernating / Low-Value' loaded from 'best_model_Hibernating_Low-Value.pth'
--- Evaluating model for Segment: Hibernating / Low-Value ---
-> Test MAE for Hibernating / Low-Value: 3.54
--- Forecasting future sales for Segment: Hibernating / Low-Value ---
--- Calculating dynamic prices for Segment: Hibernating / Low-Value ---
--- Visualizing results for Segment: Hibernating / Low-Value ---
```




```
--- Business Insights for Segment: Hibernating / Low-Value ---
1. Forecast Accuracy: The model for 'Hibernating / Low-Value' achieved a Mean Absolute Error (MAE) of 3.54 units on the test data.
   - Insight: On average, the model's daily sales forecast for this segment is off by approximately 4 units. This provides a measure of forec

2. Future Demand Outlook: Over the next 15 days, a total of 44 units are forecasted for this segment.
   - Insight: This translates to an average daily demand of 2.93 units. Use this for overall stock level planning.

3. Peak Demand: The highest forecasted demand (3 units) occurs around 2010-05-08.
   - Insight: Be prepared for a surge in demand around this date. Consider extra staffing or stocking up.

4. Low Demand: The lowest forecasted demand (3 units) occurs around 2010-04-30.
   - Insight: This period might be suitable for promotions or clearing excess inventory.

5. Dynamic Pricing: The average recommended price for this segment is $2.38.
   - Insight: This is -0.01 lower than the historical average price ($2.38). The dynamic pricing adjusts based on forecasted demand.

6. Price Variation: Recommended prices for this segment range from $2.38 to $2.39.
   - Insight: The price recommendations adapt to fluctuations in demand. Be ready to implement these price changes.

7. Estimated Financial Impact: Based on the forecast and recommended prices, the estimated total revenue is $104.43 and estimated total profi
   - Insight: This provides a potential financial outcome of implementing the dynamic pricing strategy for this segment.

8. Segment Profile: This segment ('Hibernating / Low-Value') likely has specific purchasing behaviors (e.g., price sensitivity, purchase frec
   - Insight: Consider combining these forecasts with broader segment understanding for targeted marketing or product development efforts.
```

```
--- End of Insights for Segment: Hibernating / Low-Value ---

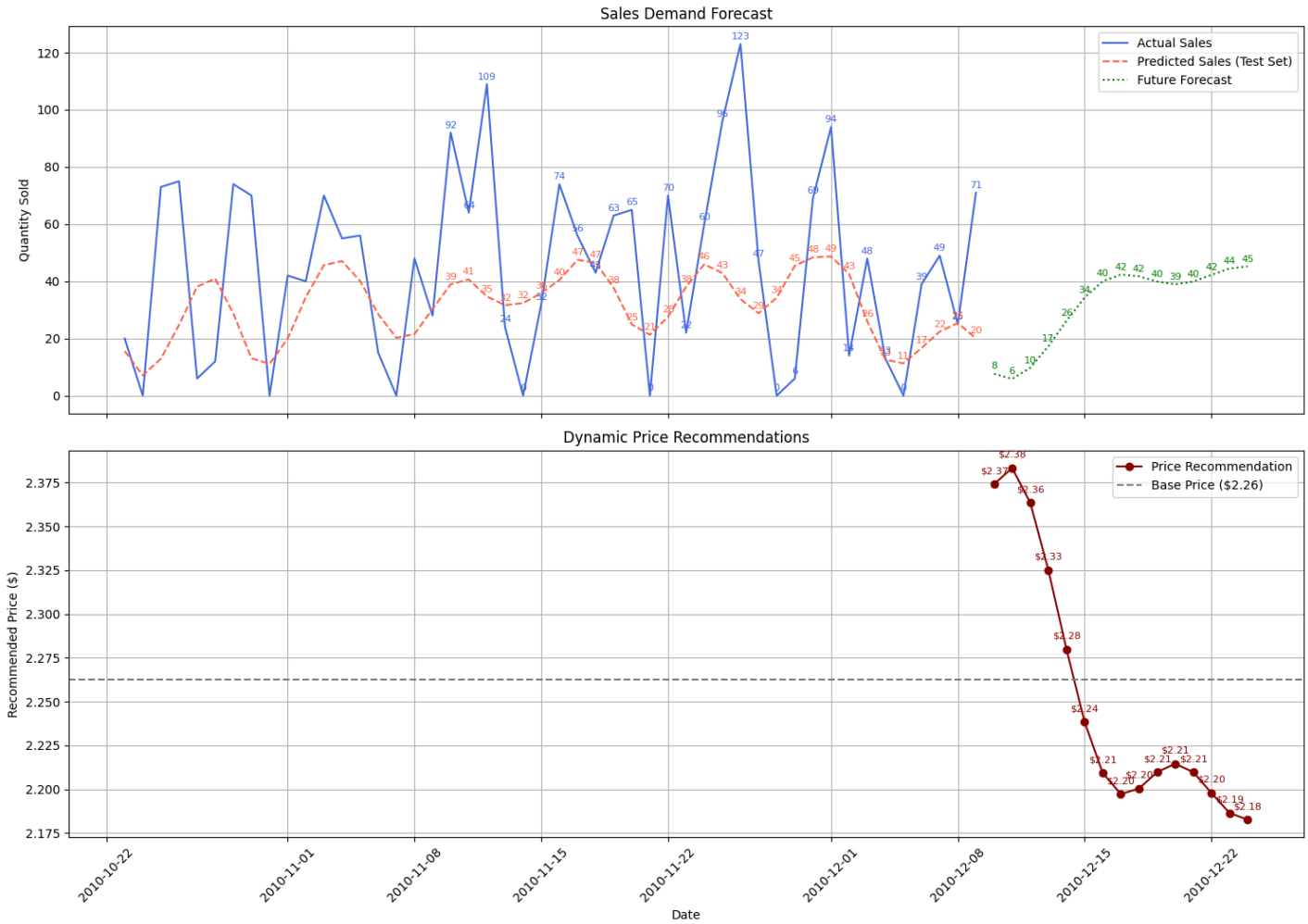
--- Future Forecast & Price Table for Hibernating / Low-Value ---
            Future_Forecast  Price_Recommendation
Date
2010-04-30             2.79                  2.39
2010-05-01             2.87                  2.38
2010-05-02             2.91                  2.38
2010-05-03             2.93                  2.38
2010-05-04             2.94                  2.38
2010-05-05             2.94                  2.38
2010-05-06             2.95                  2.38
2010-05-07             2.95                  2.38
2010-05-08             2.95                  2.38
2010-05-09             2.95                  2.38
2010-05-10             2.95                  2.38
2010-05-11             2.95                  2.38
2010-05-12             2.95                  2.38
2010-05-13             2.95                  2.38
2010-05-14             2.95                  2.38

--- Pipeline Finished for Segment: Hibernating / Low-Value ---

============================================================
--- Starting LSTM Forecast Pipeline for Product: 85099B, Segment: Potential Loyalists ---
============================================================

--- Preparing data for Segment: Potential Loyalists ---
-> Data prepared successfully for 'Potential Loyalists' with 374 daily records.
--- Training model for Segment: Potential Loyalists ---
Epoch 10/100, Val Loss: 0.0593
Epoch 20/100, Val Loss: 0.0400
Epoch 30/100, Val Loss: 0.0358
Epoch 40/100, Val Loss: 0.0481
Epoch 50/100, Val Loss: 0.0495
Early stopping at epoch 55.
-> Model for 'Potential Loyalists' loaded from 'best_model_Potential_Loyalists.pth'
--- Evaluating model for Segment: Potential Loyalists ---
-> Test MAE for Potential Loyalists: 27.01
--- Forecasting future sales for Segment: Potential Loyalists ---
--- Calculating dynamic prices for Segment: Potential Loyalists ---
--- Visualizing results for Segment: Potential Loyalists ---
```

**Forecast & Pricing for Product: 85099B (Segment: Potential Loyalists)**




```
--- Business Insights for Segment: Potential Loyalists ---
1. Forecast Accuracy: The model for 'Potential Loyalists' achieved a Mean Absolute Error (MAE) of 27.01 units on the test data.
```

```
   - Insight: On average, the model's daily sales forecast for this segment is off by approximately 27 units. This provides a measure of fore

2. Future Demand Outlook: Over the next 15 days, a total of 476 units are forecasted for this segment.
   - Insight: This translates to an average daily demand of 31.73 units. Use this for overall stock level planning.

3. Peak Demand: The highest forecasted demand (45 units) occurs around 2010-12-24.
   - Insight: Be prepared for a surge in demand around this date. Consider extra staffing or stocking up.

4. Low Demand: The lowest forecasted demand (6 units) occurs around 2010-12-11.
   - Insight: This period might be suitable for promotions or clearing excess inventory.

5. Dynamic Pricing: The average recommended price for this segment is $2.25.
   - Insight: This is -0.01 lower than the historical average price ($2.26). The dynamic pricing adjusts based on forecasted demand.

6. Price Variation: Recommended prices for this segment range from $2.18 to $2.38.
   - Insight: The price recommendations adapt to fluctuations in demand. Be ready to implement these price changes.

7. Estimated Financial Impact: Based on the forecast and recommended prices, the estimated total revenue is $1056.70 and estimated total prof
   - Insight: This provides a potential financial outcome of implementing the dynamic pricing strategy for this segment.

8. Segment Profile: This segment ('Potential Loyalists') likely has specific purchasing behaviors (e.g., price sensitivity, purchase frequenc
   - Insight: Consider combining these forecasts with broader segment understanding for targeted marketing or product development efforts.

--- End of Insights for Segment: Potential Loyalists ---

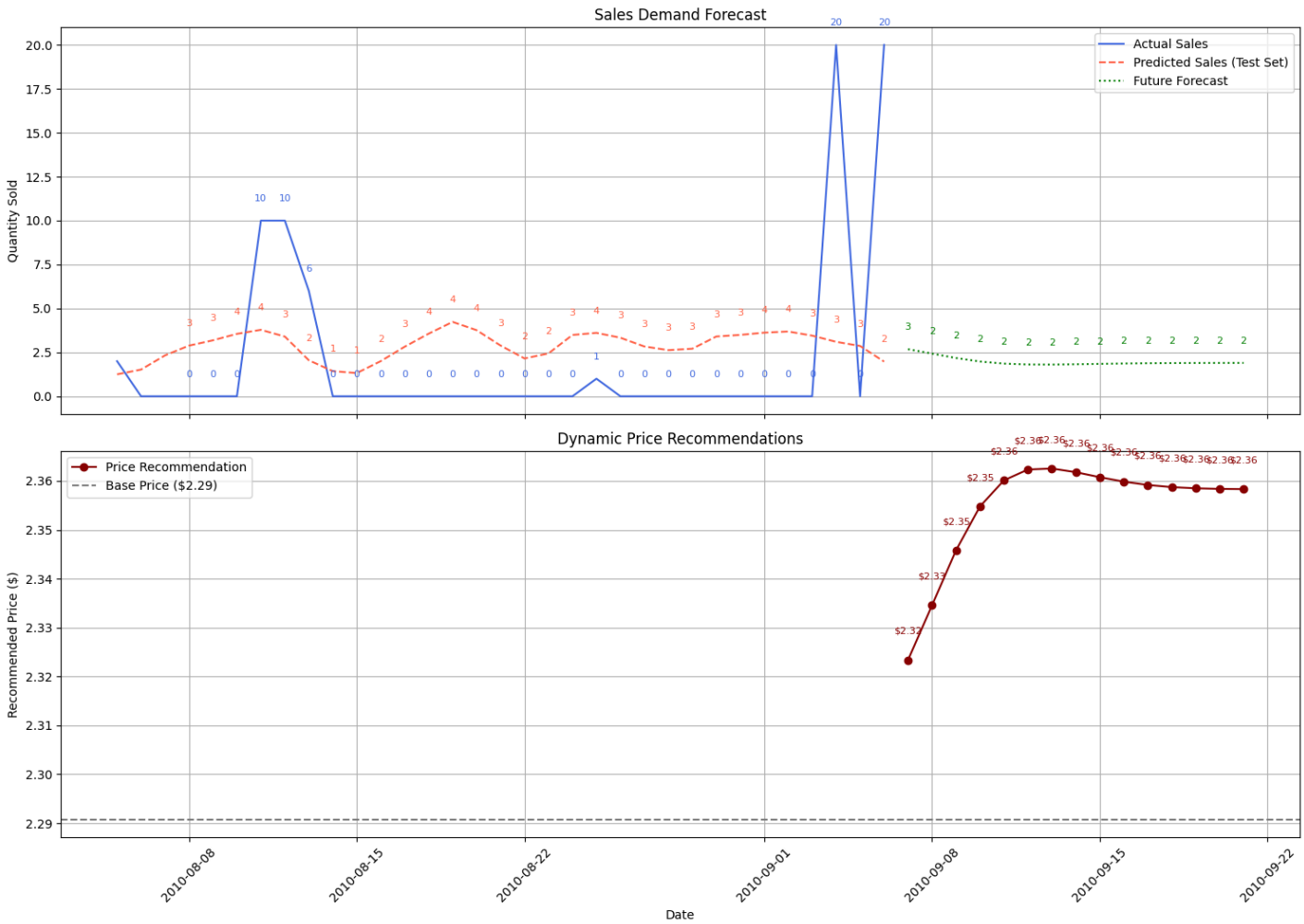
--- Future Forecast & Price Table for Potential Loyalists ---
            Future_Forecast  Price_Recommendation
Date
2010-12-10             7.67                  2.37
2010-12-11             5.86                  2.38
2010-12-12             9.76                  2.36
2010-12-13            17.29                  2.33
2010-12-14            26.16                  2.28
2010-12-15            34.30                  2.24
2010-12-16            40.01                  2.21
2010-12-17            42.36                  2.20
2010-12-18            41.75                  2.20
2010-12-19            39.92                  2.21
2010-12-20            38.97                  2.21
2010-12-21            39.91                  2.21
2010-12-22            42.24                  2.20
2010-12-23            44.48                  2.19
2010-12-24            45.23                  2.18

--- Pipeline Finished for Segment: Potential Loyalists ---

============================================================
--- Starting LSTM Forecast Pipeline for Product: 85099B, Segment: At-Risk Customers ---
============================================================

--- Preparing data for Segment: At-Risk Customers ---
-> Data prepared successfully for 'At-Risk Customers' with 277 daily records.
--- Training model for Segment: At-Risk Customers ---
Epoch 10/100, Val Loss: 0.0663
Epoch 20/100, Val Loss: 4.3388
Epoch 30/100, Val Loss: 0.0382
Epoch 40/100, Val Loss: 0.0541
Early stopping at epoch 46.
-> Model for 'At-Risk Customers' loaded from 'best_model_At-Risk_Customers.pth'
--- Evaluating model for Segment: At-Risk Customers ---
-> Test MAE for At-Risk Customers: 3.96
--- Forecasting future sales for Segment: At-Risk Customers ---
--- Calculating dynamic prices for Segment: At-Risk Customers ---
--- Visualizing results for Segment: At-Risk Customers ---
```

**Forecast & Pricing for Product: 85099B (Segment: At-Risk Customers)**




```
--- Business Insights for Segment: At-Risk Customers ---
1. Forecast Accuracy: The model for 'At-Risk Customers' achieved a Mean Absolute Error (MAE) of 3.96 units on the test data.
   - Insight: On average, the model's daily sales forecast for this segment is off by approximately 4 units. This provides a measure of forec

2. Future Demand Outlook: Over the next 15 days, a total of 30 units are forecasted for this segment.
   - Insight: This translates to an average daily demand of 1.98 units. Use this for overall stock level planning.

3. Peak Demand: The highest forecasted demand (3 units) occurs around 2010-09-07.
   - Insight: Be prepared for a surge in demand around this date. Consider extra staffing or stocking up.

4. Low Demand: The lowest forecasted demand (2 units) occurs around 2010-09-13.
   - Insight: This period might be suitable for promotions or clearing excess inventory.

5. Dynamic Pricing: The average recommended price for this segment is $2.35.
   - Insight: This is 0.06 higher than the historical average price ($2.29). The dynamic pricing adjusts based on forecasted demand.

6. Price Variation: Recommended prices for this segment range from $2.32 to $2.36.
   - Insight: The price recommendations adapt to fluctuations in demand. Be ready to implement these price changes.

7. Estimated Financial Impact: Based on the forecast and recommended prices, the estimated total revenue is $69.95 and estimated total profit
   - Insight: This provides a potential financial outcome of implementing the dynamic pricing strategy for this segment.

8. Segment Profile: This segment ('At-Risk Customers') likely has specific purchasing behaviors (e.g., price sensitivity, purchase frequency)
   - Insight: Consider combining these forecasts with broader segment understanding for targeted marketing or product development efforts.

--- End of Insights for Segment: At-Risk Customers ---

--- Future Forecast & Price Table for At-Risk Customers ---
            Future_Forecast  Price_Recommendation
Date
2010-09-07             2.68                  2.32
2010-09-08             2.43                  2.33
2010-09-09             2.18                  2.35
2010-09-10             1.98                  2.35
2010-09-11             1.86                  2.36
2010-09-12             1.81                  2.36
2010-09-13             1.80                  2.36
2010-09-14             1.82                  2.36
2010-09-15             1.84                  2.36
2010-09-16             1.86                  2.36
2010-09-17             1.88                  2.36
2010-09-18             1.89                  2.36
2010-09-19             1.89                  2.36
2010-09-20             1.90                  2.36
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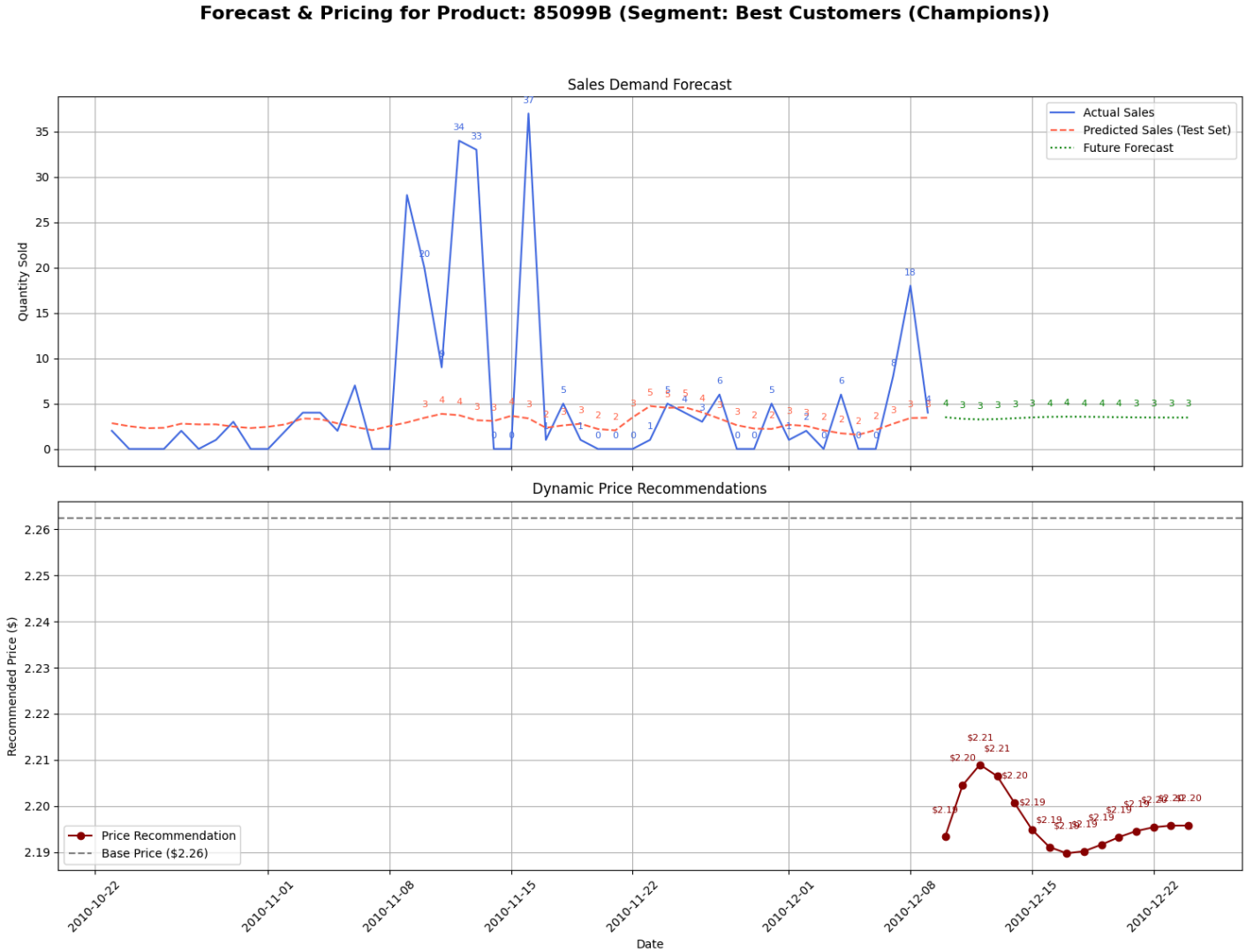
```
2010-09-21             1.90                  2.36

--- Pipeline Finished for Segment: At-Risk Customers ---

============================================================
--- Starting LSTM Forecast Pipeline for Product: 85099B, Segment: Best Customers (Champions) ---
============================================================

--- Preparing data for Segment: Best Customers (Champions) ---
-> Data prepared successfully for 'Best Customers (Champions)' with 374 daily records.
--- Training model for Segment: Best Customers (Champions) ---
Epoch 10/100, Val Loss: 0.0111
Epoch 20/100, Val Loss: 0.0108
Epoch 30/100, Val Loss: 0.0113
Epoch 40/100, Val Loss: 0.0133
Epoch 50/100, Val Loss: 0.0148
Early stopping at epoch 50.
-> Model for 'Best Customers (Champions)' loaded from 'best_model_Best_Customers_(Champions).pth'
--- Evaluating model for Segment: Best Customers (Champions) ---
-> Test MAE for Best Customers (Champions): 4.99
--- Forecasting future sales for Segment: Best Customers (Champions) ---
--- Calculating dynamic prices for Segment: Best Customers (Champions) ---
--- Visualizing results for Segment: Best Customers (Champions) ---
```




```
--- Business Insights for Segment: Best Customers (Champions) ---
1. Forecast Accuracy: The model for 'Best Customers (Champions)' achieved a Mean Absolute Error (MAE) of 4.99 units on the test data.
   - Insight: On average, the model's daily sales forecast for this segment is off by approximately 5 units. This provides a measure of forec

2. Future Demand Outlook: Over the next 15 days, a total of 52 units are forecasted for this segment.
   - Insight: This translates to an average daily demand of 3.45 units. Use this for overall stock level planning.

3. Peak Demand: The highest forecasted demand (4 units) occurs around 2010-12-17.
   - Insight: Be prepared for a surge in demand around this date. Consider extra staffing or stocking up.

4. Low Demand: The lowest forecasted demand (3 units) occurs around 2010-12-12.
   - Insight: This period might be suitable for promotions or clearing excess inventory.

5. Dynamic Pricing: The average recommended price for this segment is $2.20.
   - Insight: This is -0.07 lower than the historical average price ($2.26). The dynamic pricing adjusts based on forecasted demand.

6. Price Variation: Recommended prices for this segment range from $2.19 to $2.21.
   - Insight: The price recommendations adapt to fluctuations in demand. Be ready to implement these price changes.

7. Estimated Financial Impact: Based on the forecast and recommended prices, the estimated total revenue is $113.74 and estimated total profi
   - Insight: This provides a potential financial outcome of implementing the dynamic pricing strategy for this segment.
```

```
8. Segment Profile: This segment ('Best Customers (Champions)') likely has specific purchasing behaviors (e.g., price sensitivity, purchase f
   - Insight: Consider combining these forecasts with broader segment understanding for targeted marketing or product development efforts.

--- End of Insights for Segment: Best Customers (Champions) ---

--- Future Forecast & Price Table for Best Customers (Champions) ---
            Future_Forecast  Price_Recommendation
Date
2010-12-10             3.50                  2.19
2010-12-11             3.32                  2.20
2010-12-12             3.25                  2.21
2010-12-13             3.29                  2.21
2010-12-14             3.39                  2.20
2010-12-15             3.48                  2.19
2010-12-16             3.54                  2.19
2010-12-17             3.56                  2.19
2010-12-18             3.55                  2.19
2010-12-19             3.53                  2.19
2010-12-20             3.50                  2.19
2010-12-21             3.48                  2.19
2010-12-22             3.47                  2.20
2010-12-23             3.46                  2.20
2010-12-24             3.46                  2.20

--- Pipeline Finished for Segment: Best Customers (Champions) ---

======================================================================
---           OVERALL BUSINESS INSIGHTS SUMMARY           ---
======================================================================

Analysis for Product Code: 85099B

🎯 Most Accurate Forecast: 'Hibernating / Low-Value'
   - Model for this segment has the lowest error (MAE: 3.54), making its forecast the most reliable.

📈 Highest Demand Segment: 'Potential Loyalists'
   - This group is predicted to buy the most units (476 units) in the next 15 days.
   - STRATEGY: Ensure sufficient stock for this segment and consider targeted marketing.

💰 Highest Price Tolerance: 'Hibernating / Low-Value'
   - This group's demand forecast supports the highest average price ($2.38).
   - STRATEGY: Focus premium product versions or bundles on this segment.

📦 Total Forecasted Demand (All Processed Segments):
   - A total of 601 units are predicted to be sold in the next 15 days across segments with sufficient data.

📊 Insights from Combined Plot:
   - VISUAL ANALYSIS: The combined forecast plot shows that the demand for 'Potential Loyalists' is consistently higher than other segments.
   - PRICING DYNAMICS: The recommended prices for 'Hibernating / Low-Value' are visibly higher, confirming their lower price sensitivity.

======================================================================
```